

Basics of Electronics Engineering

Aaryan Khedekar

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1 Diodes

1.1 Semiconductors:

1.1.1 Types of Materials:

1. Insulators: Band Gap of 7.5eV.
2. Semiconductors: Band Gap of 1eV.
3. Conductors: Band Gap of 0eV.

1.1.2 Intrinsic Semiconductors:

1. They are pure Silicon or Germanium Semiconductors.
2. They have equal number of Electrons and Holes.
3. They form covalent bonds between each other and are stable.
4. At 0K, they have no free Electrons or Holes.
5. At room temperature, there exists free electrons, forming holes, thereby attracting electrons from nearby atoms inside the semiconductor.
6. **Intrinsic Concentration:** At room temperature,

$$n_i = n_h = n_e = 1.5 \times 10^{10}/cm^3$$

1.1.3 Extrinsic Semiconductors:

1. **Doping:** Adding impurities to the semiconductor to increase the number of free electrons or holes.
2. **Pentavalent Dopings:**
 - (a) Addition of impurities like Phosphorus, Arsenic (Group 15 elements).
 - (b) They give additional electrons to the semiconductor.

(c) **Donor Concentration:**

$$N_D = n_e = 1.5 \times 10^{17}/cm^3$$

(d) **Hole Concentration:**

$$n_h = \frac{n_i^2}{n_e} = 1.5 \times 10^3/cm^3$$

(e) Here, electron concentration increases and hole concentration decreases. Hence, majority carriers are electrons and minority carriers are holes.

(f) This type of semiconductor is called **N-type** semiconductor.

3. Trivalent Dopings:

(a) Addition of impurities like Boron, Aluminium (Group 13 elements).

(b) They give additional holes to the semiconductor.

(c) **Acceptor Concentration:**

$$N_A = n_h = 1.5 \times 10^{17}/cm^3$$

(d) **Electron Concentration:**

$$n_e = \frac{n_i^2}{n_h} = 1.5 \times 10^3/cm^3$$

(e) Here, hole concentration increases and electron concentration decreases. Hence, majority carriers are holes and minority carriers are electrons.

(f) This type of semiconductor is called **P-type** semiconductor.

4.

$$n_i = BT^{3/2}e^{-E_g/2kT}$$

where

(a) B is the "material constant" $B_{silicon} = 7.2 \times 10^{15}/cm^3/K^{3/2}$

(b) T is the temperature in Kelvin.

(c) E_g is the band gap energy.

(d) k is the Boltzmann constant.

1.2 Current Flow:

1. **Drift Current:** Current due to the flow of charge carriers under the influence of an electric field.

$$v_d \propto E \implies v_d = \mu_e E$$

Now,

$$I = \frac{Q}{t} = \frac{n_e q_e A l}{t} = n_e A q_e v_d = n_e A q_e \mu_e E$$

Hence,

$$J = n_e q_e \mu_e E \implies J = \sigma E$$

Due to both holes and electrons,

$$J = q(n_e \mu_e + n_h \mu_h) E$$

where $q = -q_e = q_h$ and μ denotes the mobility.

2. **Diffusion Current:** Current due to the flow of charge carriers from a region of high concentration to a region of low concentration.

$$J_p = -q D_p \frac{dp}{dx} \text{ and } J_n = q D_n \frac{dn}{dx}$$

where D is the diffusion constant.

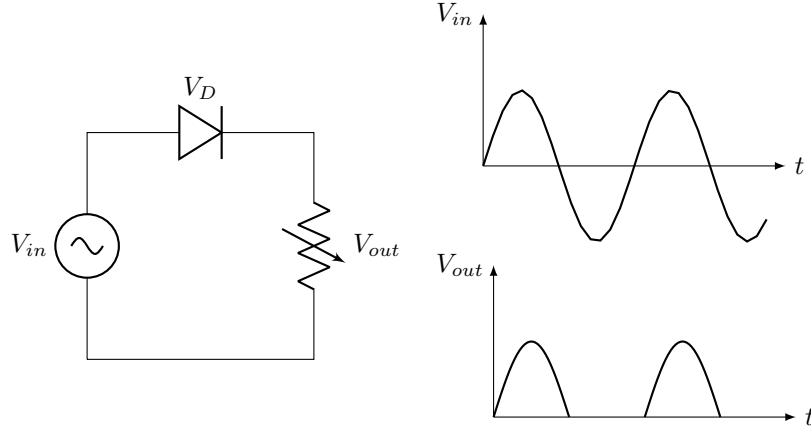
$$\frac{D_p}{\mu_p} = \frac{D_n}{\mu_n} = \frac{kT}{q} = V_T \text{ (thermal voltage)}$$

1.3 P-N Junction Diodes:

1. **P-N Junction:** A junction formed by joining P-type and N-type semi-conductors.
2. **Depletion Region:** The region near the junction where the charge carriers are depleted.
3. **Barrier Potential:** The potential difference across the junction.
4. **Forward Bias:** Applying a positive voltage to the P-type and negative voltage to the N-type. This reduces the barrier potential.
5. **Reverse Bias:** Applying a negative voltage to the P-type and positive voltage to the N-type. This increases the barrier potential.
6. **Knee voltage:** The voltage at which the diode starts conducting.
7. **Breakdown Voltage:** The voltage at which the diode breaks down and conducts.
8. **Reverse Saturation Current:** The current that flows when the diode is reverse biased.

1.4 Half Wave Rectifier:

1. The below circuit is called a "half-wave rectifier".



- 2.

$$V_{avg} = V_{DC} = \frac{1}{T} \int_0^T V_{in}(t) dt = \frac{V_m}{\pi}$$

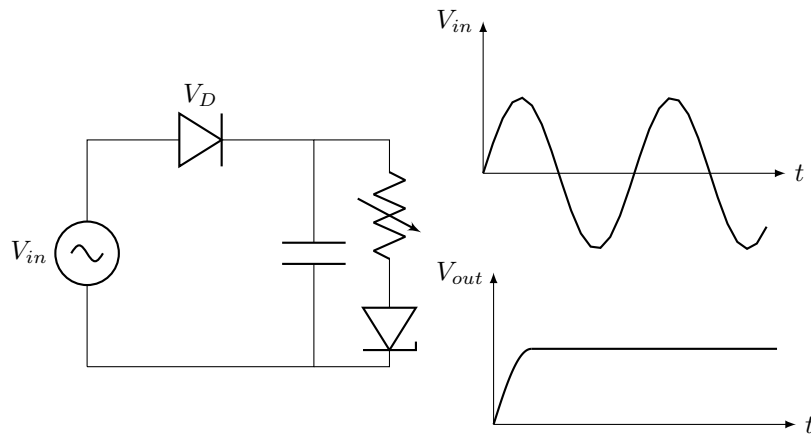
- 3.

$$V_{rms} = \frac{V_m}{2}$$

4. Ripple factor = $\frac{V_{AC}}{V_{DC}}$ and we know that $V_{rms}^2 = V_{AC}^2 + V_{DC}^2$

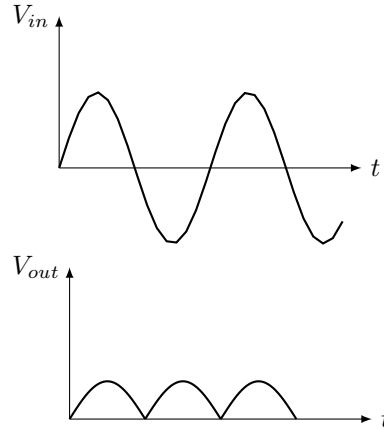
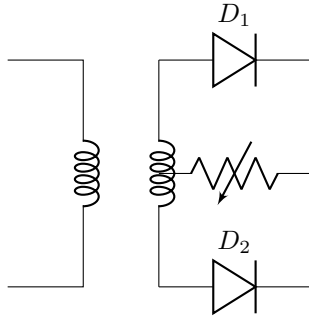
$$RF = \frac{\sqrt{V_{rms}^2 - V_{DC}^2}}{V_{DC}} = \sqrt{\left(\frac{V_{rms}}{V_{DC}}\right)^2 - 1} = \sqrt{\left(\frac{V_m/2}{V_m/\pi}\right)^2 - 1} = 1.211$$

5. On adding a capacitor and a zener diode, the voltage is more regulated.



1.5 Full Wave Rectifier:

1. The below circuit is called a "full-wave rectifier".



1.6 Diode Equation:

$$I_D = I_0(e^\alpha - 1) \text{ where } \alpha = \frac{V}{\eta V_T}$$

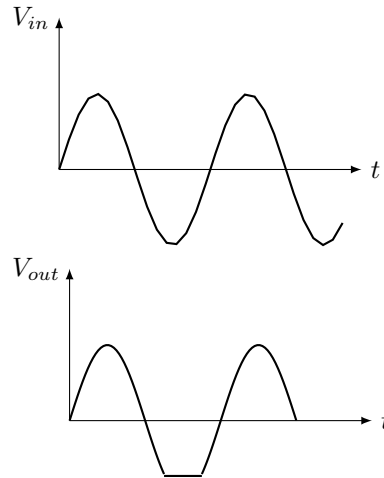
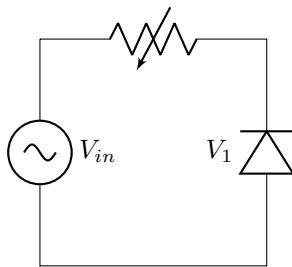
where V_T = thermal voltage

η = ideality factor (for normal diodes, $\eta = 1$)

I_0 = reverse saturation current

1.7 Clipper Circuit:

1. The below circuit is called a "clipper circuit".

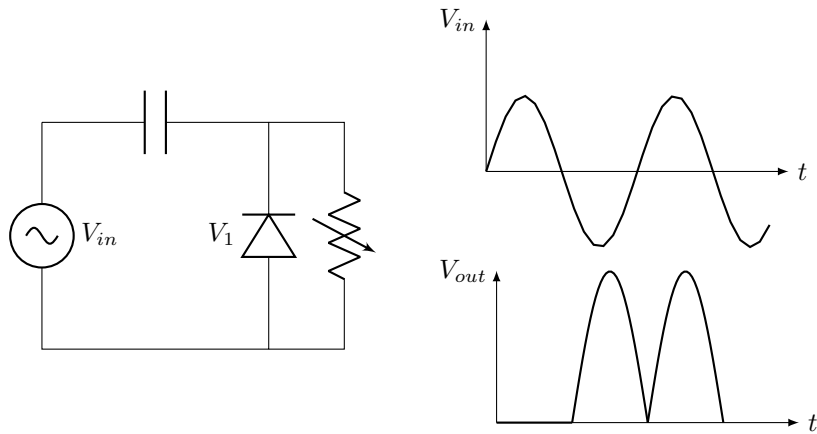


2. As this is clipping in the negative half cycle, it is called as a "negative clipper".

3. To make a positive clipper, the diode should be reversed.

1.8 Clamper Circuit:

1. The below circuit is called a "clamper circuit".



2. As this is clamping the positive half cycle, it is called as a "positive clamper".
3. To make a negative clamper, the diode should be reversed.

2 Transistors

2.0.1 Types of Transistors:

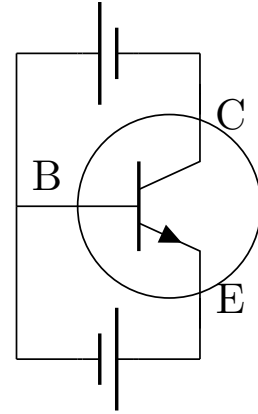
1. BJT (Bipolar Junction Transistor)
2. VJT (Vertical Junction Transistor)
3. MOSFET (Metal Oxide Semiconductor Field Effect Transistor)

2.1 Bipolar Junction Transistor:

1. These consist of three sections: Emitter, Base and Collector.
2. The emitter is heavily doped, the base is lightly doped and the collector is moderately doped.
3. There are two junctions: Emitter-Base and Base-Collector.
4. The emitter is forward biased and the collector is reverse biased.
5. There are two types of BJTs: npn and pnp.
6. In npn BJT, Emitter and Collector are n-type and Base is p-type.
7. In pnp BJT, Emitter and Collector are p-type and Base is n-type.

2.1.1 Common Base npn:

1. EB Junction is in forward bias whereas CB Junction is in reverse bias.
2. Due to forward bias high number of electrons passed through the EB junction.
3. Due to the high concentration in P region they move towards the collector side.
4. In this process few electrons recombine with the holes in base region.
5. The loss of holes in the P region is supplied by external voltage source.
6. The electrons reaching the BC Junction are swept out immediately.

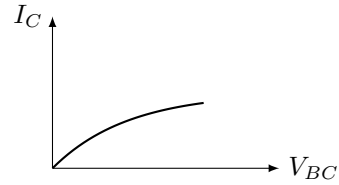
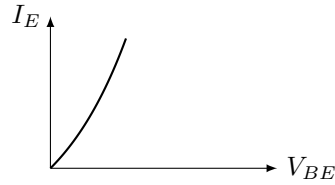


Collector Current, $I_C = \alpha I_E$
 By KCL, $I_E = I_C + I_B \implies I_E = \alpha I_E + I_B$

$$I_E(1 - \alpha) = I_B \implies \frac{I_B}{I_E} = 1 - \alpha$$

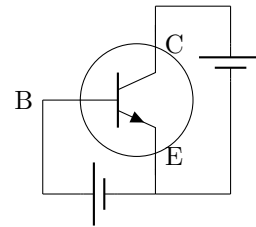
Also, $I_C = \beta I_B$.

where β is called the collector current gain.



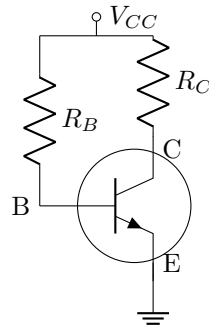
2.1.2 Common Emitter npn:

1. As I_B increase, I_C also increase, making the BC Junction more forward biased.
2. **Active Region:** The region where the transistor is used as an amplifier.
3. **Saturation Region:** The region where the transistor is used as a switch.
4. **Cut-off Region:** The region where the transistor is off.



2.1.3 Biasing:

Fixed Bias Configuration:



We know, $V_{BE} = 0.7V$.

$$I_B = \frac{V_{cc} - 0.7}{R_B}, I_C = \beta I_B$$

And $V_{CE} = V_{cc} - I_C R_C$

2.2 Field Effect Transistor:

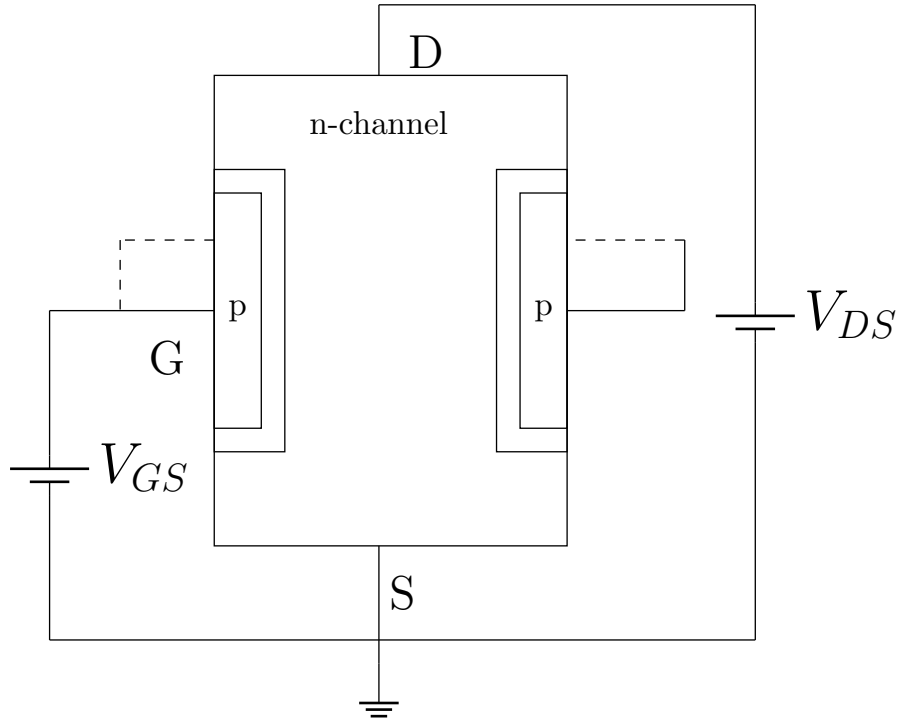
They are of two types:

1. Junction FET (JFET)
2. Metal Oxide Semiconductor FET (MOSFET)
 - (a) Enhancement Type
 - (b) Depletion Type

Bipolar Junction Transistor (BJT)	Field Effect Transistor (FET)
It is a current control device.	It is a voltage control device.
Current is due to both electron and holes.	Current is due to only one charge carrier.
It is highly temperature dependent.	It is lowly temperature dependent.
It has low input impedance.	It has high input impedance.
Current Gain is high.	Current Gain is moderate.

2.2.1 Junction FET (JFET):

1. It is a unipolar device.
2. It has three terminals:
Gate ($\equiv$ Base), Source ($\equiv$ Emitter) and Drain ($\equiv$ Collector).
3. It is of two types: N-channel and P-channel.

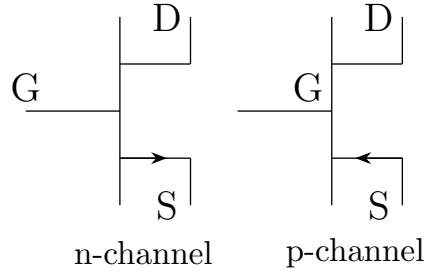


4. If V_{GS} is negative and V_{DS} is positive, the JFET is in the saturation region.
5. It means that both are in reverse bias and the junction width slowly increases. Hence the n-channel width decreases, and hence the current decreases.

6. **Equation:**

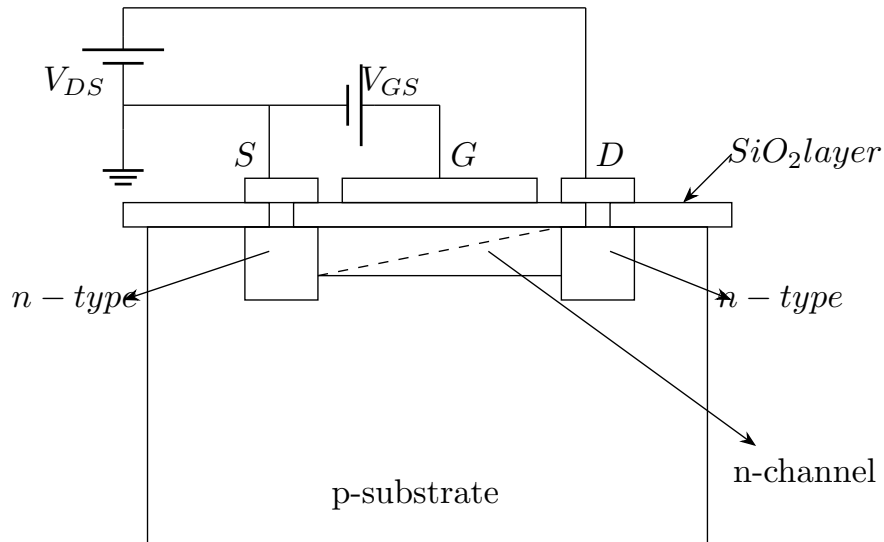
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2$$

7. **Symbols:**



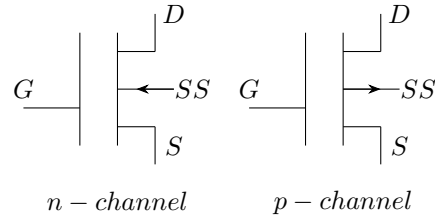
2.2.2 Depletion Type MOSFET:

1. The below is figure of a Depletion Type MOSFET:



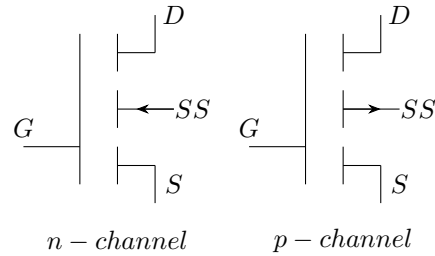
2. The characteristics of a Depletion Type MOSFET are same as that of the JFET.
3. The only difference is that the MOSFET has a thin layer of SiO_2 between the gate and the channel.
4. Channel width is decreased due to electrons being pushed downwards.

5. The below are the symbols for the Depletion Type MOSFET:



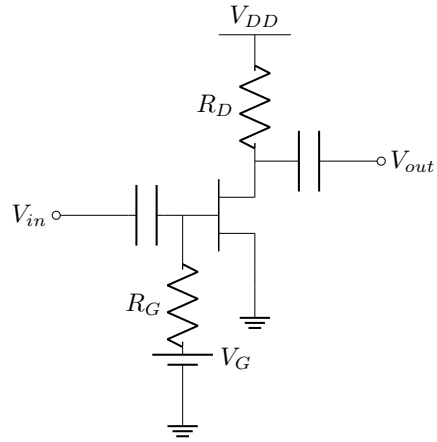
2.2.3 Enhancement Type MOSFET:

1. It is same as the above type, but the channel is not present initially.
2. The channel is created by applying a positive voltage to the gate terminal.
3. For the channel to be created, the gate voltage should be greater than a threshold voltage ($V_{GS} > V_T$).
4. **Equation:** $I_D = k(V_{GS} - V_T)^2$ where k is a transistor parameter.
5. For saturation, $V_{DS} \geq V_{GS} - V_T$
6. The below are the symbols for the Enhancement Type MOSFET:



2.2.4 Biasing:

Fixed Biasing:

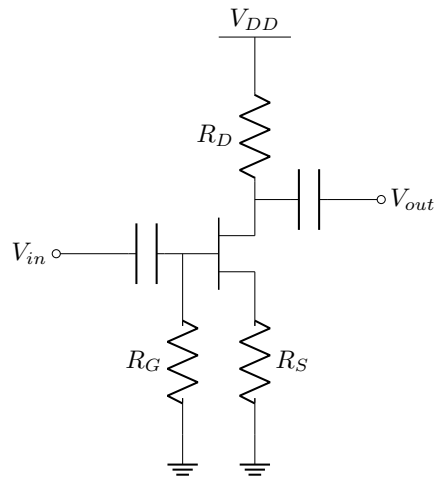


$$V_{GS} = V_G - V_S = V_G$$

$$I_D = I_{DSS} \left(1 + \frac{V_G}{V_p} \right)^2$$

$$V_{DD} = I_D R_D + V_{DS}$$

Self Biasing:



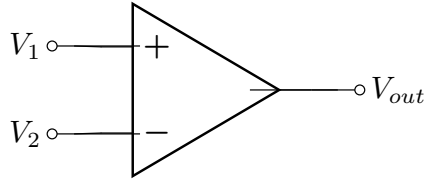
$$V_G = 0, V_S = I_D R_S$$

$$V_{GS} = V_G - V_S = -I_D R_S$$

$$I_D = I_{DSS} \left(1 + \frac{I_D R_S}{V_p} \right)^2$$

3 Operational Amplifiers:

3.1 Introduction:



A is called as the "amplifier gain"

$$V_{out} = A(V_1 - V_2)$$

$$(V_{out})_{max} = +V_{cc} = +V_{sat}$$

$$(V_{out})_{min} = -V_{cc} = -V_{sat}$$

3.1.1 Desired Characteristics of an Operational Amplifier:

1. It should have **high gain**.
2. It should have **high input impedance**.
3. It should have **low output impedance**.
4. It should have **high bandwidth**.

3.1.2 Classification of Operational Amplifiers:

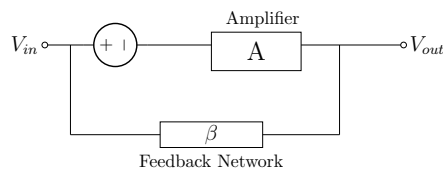
1. Amplifiers:

- (a) They have (-ve) feedback.
- (b) Applications include rectifiers, clippers, clampers, etc.

2. Comparators:

- (a) They have (+ve) feedback.
- (b) Applications include waveform generator, etc.

3.1.3 Amplifiers as feedback network:



Advantages of Feedback Network:

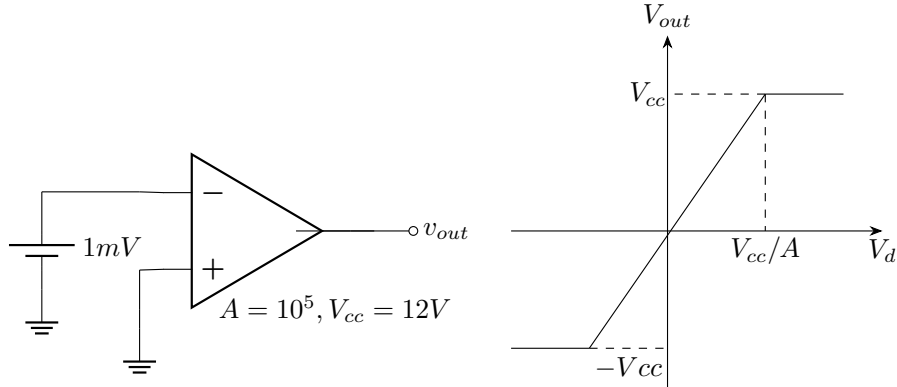
1. Gain Control.
2. Output values are less susceptible to changes in gain (A)

$$V_{out} = A(V_{in} + \beta V_{out}) \implies V_{out}(1 + A\beta) = AV_{in}$$

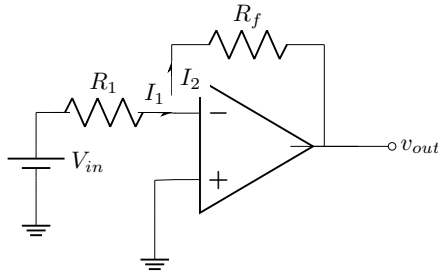
$$\therefore V_{out} = \frac{A}{1 + A\beta} V_{in} \text{ and if } A \rightarrow \infty \implies V_{out} = \frac{1}{\beta} V_{in}$$

3.2 OPAMPs as Amplifiers:

3.2.1 Open Loop OPAMPs:



3.2.2 Inverting Amplifiers:



1. Input is applied to the inverting terminal.
2. Output is 180 degrees out of phase with input.
3. As current entering opamp is really small, $I_1 = I_2$.

$$V_{out} = A(0 - V_a) \implies V_a = -\frac{1}{A}V_{out}$$

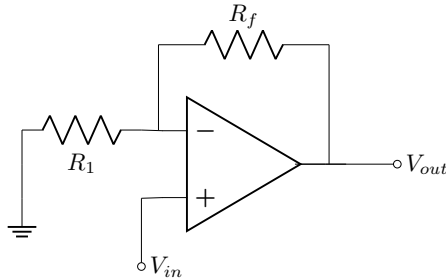
For ideal case, $A \rightarrow \infty \implies V_a = 0$

$$\therefore I_1 = \frac{V_{in}}{R_1} \implies I_2 = \frac{V_{in}}{R_1} = \frac{-V_{out}}{R_f}$$

$$\therefore A_v = -\frac{R_f}{R_1} \implies V_{out} = -\frac{R_f}{R_1}V_{in}$$

Virtual Ground: The voltage at the inverting terminal is zero.

3.2.3 Non-Inverting Amplifiers:

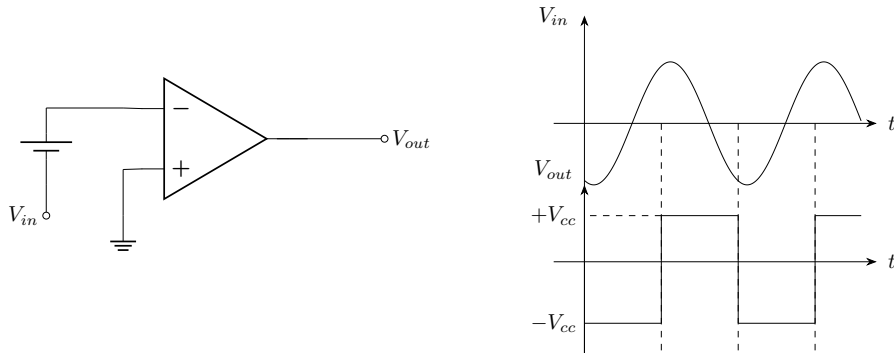


1. Input is applied to the non-inverting terminal.
2. Output is in phase with input.
3. As V_{out} is finite and A is very large, $V^+ = V^- = V_{in}$

By KCL,

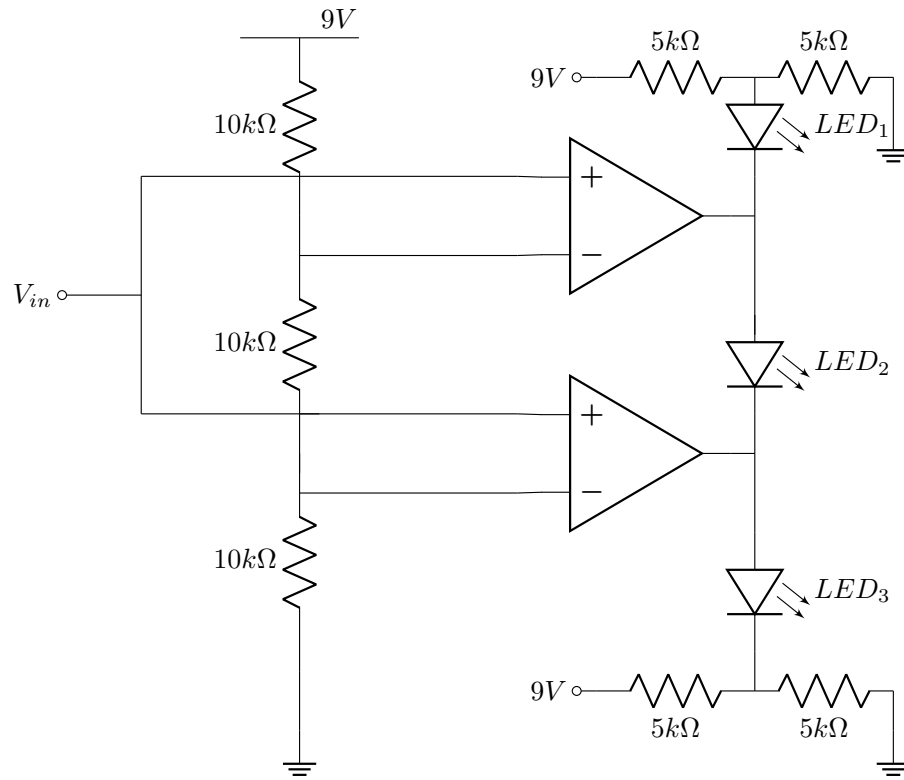
$$\frac{V_{in} - 0}{R_1} = \frac{V_{in} - V_{out}}{R_f} \Rightarrow V_{out} = \left(1 + \frac{R_f}{R_1}\right) V_{in}$$

3.3 OPAMPs as Comparators:



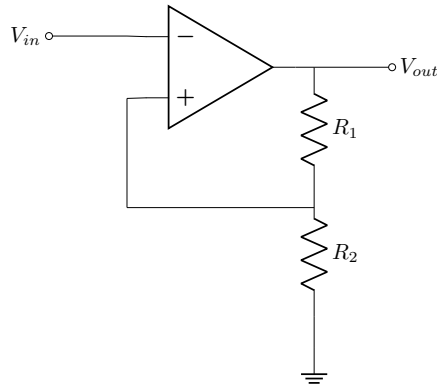
They can be used as a "zero-crossing detector".

3.3.1 Voltage Level Identifier:



V Range	V_{O1}	V_{O2}	LED_1	LED_2	LED_3
0-3V	$+V_{cc}$	$+V_{cc}$	OFF	OFF	ON
3-6V	$-V_{cc}$	$+V_{cc}$	OFF	ON	OFF
6-9V	$-V_{cc}$	$-V_{cc}$	ON	OFF	OFF

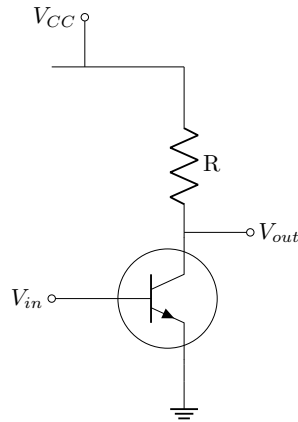
3.3.2 Schmitt Trigger:



$$V_{out} = V_{cc} \left(\frac{R_2}{R_1 + R_2} \right)$$

4 Digital Electronics

4.0.1 BJT as NOT Gate:



1. If $V_{in} < 0.6 \Rightarrow I_B = 0 \Rightarrow V_{out} \approx V_{cc}$.
2. If $V_{in} \gg 0.6 \Rightarrow I_B$ is large $\Rightarrow I_C$ is very large.
Now, $(V_{out})_{max} = V_{CC} \Rightarrow (I_C)_{max} = V_{CC}/R \Rightarrow V_{out} \approx 0$.
3. Hence, this circuit can be used as a "NOT Gate".
4. In active region, $I_C = \beta I_B$ is followed, hence amplification occur.
5. In saturation and cutoff regions, $I_C \neq \beta I_B$, hence switching occur.

4.1 Number Systems:

4.1.1 Base or Radix

The number of digits in a number system is called its base or radix.

1. Binary: Base 2 (0, 1)
2. Octal: Base 8 (0-7)
3. Decimal: Base 10 (0-9)
4. Hexadecimal: Base 16 (0-9, A-F)

If a_i are digits, then a number of base "b" can be written as:

$$(a_{N-1}a_{N-2}\dots a_1a_0)_b = a_{N-1}b^{N-1} + a_{N-2}b^{N-2} + \dots + a_1b^1 + a_0b^0$$

4.1.2 Converting Decimal to Binary:

1. For digits before decimal point, repeatedly divide the number by 2 and the equivalent binary number is obtained by writing the remainders in reverse order.
2. For digits after decimal point, repeatedly multiply the number by 2 and the equivalent binary number is obtained by writing the integer part of the product.

4.1.3 Complements:

For a number "N" with "n" digits with base "b",

1. Radix Complement: $C_r = b^n - N$
2. Diminished Radix Complement: $C_d = (b^n - 1) - N$

4.1.4 Subtraction of two numbers:

1. Consider two base "b" numbers M and N, then the subtraction M - N can be calculated by taking the radix complement of N and adding it to M.
2. If a carry bit is generated, the result is positive and can be obtained by omitting the carry bit.
3. If it does not generate, the result is negative and can be obtained by taking the complement of the result.

4.1.5 Magnitude Representation:

1. **Signed Magnitude Representation:** For a 8 bit binary number, the MSB represents the sign (0 = positive, 1 = negative) and rest of the bits determine the magnitude of the number.
2. **Unsigned Magnitude Representation:** For a 8 bit binary number all eight bits describe the magnitude.

4.1.6 BCD Numbers:

These are called as "Binary Coded Decimal" numbers and all the decimal digits are represented in Binary.

4.1.7 Codes:

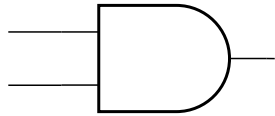
Decimal	8421	Excess-3	Graycode
0	0000	0011	0000
1	0001	0100	0001
2	0010	0101	0011
3	0011	0110	0010
4	0100	0111	0110
5	0101	1000	0111
6	0110	1001	0101
7	0111	1010	0100
8	1000	1011	1100
9	1001	1100	1101

1. Excess-3 is a self complementary code.
2. Each digit in graycode is one bit different from the previous digit. Each digit is XORed with the preceeding digit and the MSB is unaltered.

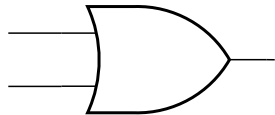
4.2 Boolean Functions:

1. A Boolean function is a function that takes binary inputs and returns binary outputs.
2. The basic operations in Boolean algebra are AND, OR, NOT.
3. For n variables, 2^n possible combinations can be formed.

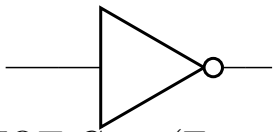
4.2.1 Logic Gates:



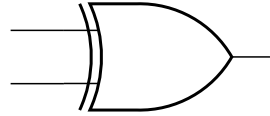
AND Gate ($F = x \cdot y$)



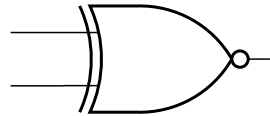
OR Gate ($F = x + y$)



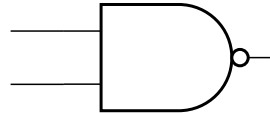
NOT Gate ($F = \bar{x}$)



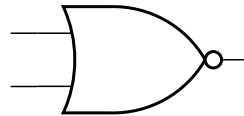
XOR Gate ($F = x \oplus y$)



XNOR Gate ($F = x \bar{\oplus} y$)



NAND Gate ($F = x \bar{\cdot} y$)



NOR Gate ($F = x \bar{+} y$)